OS Concepts and System Programming in Linux

→ Kernel Features of OS

- → CPU mgt service
- → Memory mgt service
- → File mgt service
- → N/W mgt "
- → IPC & Synch
- → etc..

utilise these classes as Q&A sessions.

→ helps in anticipating next question

→ You think like an interviewer.

→ Can OS be present without kernel ?

→ Impossible

→ Can kernel be present without being known as OS ?

- yes / Partially No

→ We may create/configure a kernel without apm.

→

1st week : PCB, States, Job creation, Signals, Multitasking and multithreading } Process mgt

2nd week : File mgt & File Handling

3rd week : IPC & Synchronisation

4th week : Misc.

What is a Process in Linux ?

→ App under exec. is a Process.

How to know the list of jobs (processes) currently Active ?

→ System monitor, top, ps

```
$ps -e | wc -l ↵
286
$ ps -e | wc -l ↵
304
$ ps -eL | wc -l ↵
703
```

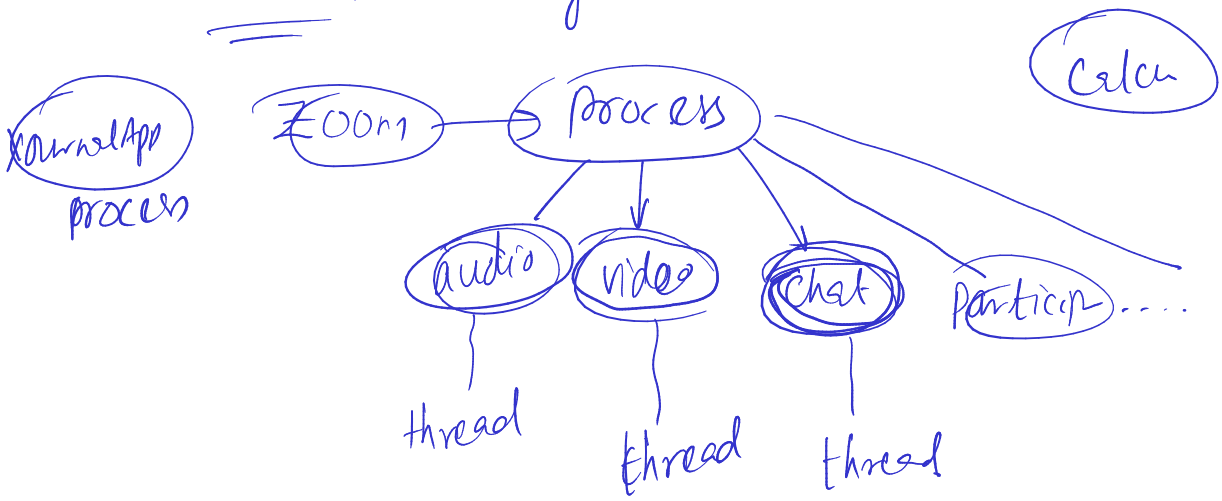
→ no. of processes

→ No. of threads

```
$./a.out ↵
MENU:
i> input
d> del
:
Enter your ch :
```

Zoom App launched.

→ Zoom in Memory



If Zoom like app, uses each of its subtasks (like chat, participant, etc..) as process, then what happens?

- Memory overhead, time overhead

→ context switching time increases, hence cpu-performance degrades.

→ Additional mem-Area (sep. pages) are used.

$ps -e ↵

304

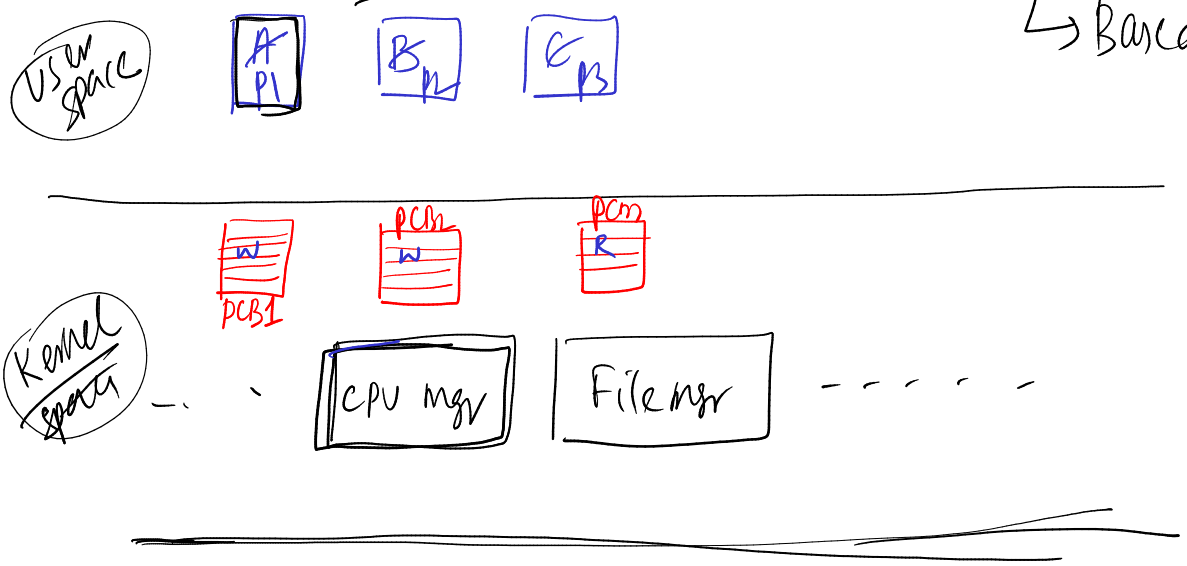
$

Although, list of processes 304,

All the 304 are NOT scheduled for CPU.

Because, All the jobs are not requiring CPU. Few jobs may be in delay state / wait state / suspended / etc.

- CPU-scheduler (part of taskmgr) is responsible for separation of jobs — which need CPU and jobs which NOT.

- It is achieved by state of the process

→ Only Ready State jobs are scheduled for CPU.
↳ Based on Sched. algo